

11 · Ad exposure is self-selected — the true effect (instrumental variables · CausalPy)

July 9, 2026

1 11 · Ad exposure is self-selected — the true effect (instrumental variables · CausalPy)

Runs in the legacy environment (pymc<6): `make env-legacy, kernel cmp-legacy.`

The business decision. People who see our ads convert more — but the customers who *end up* seeing our ads are also the ones already leaning toward us (they browse, they search, they get retargeted). So **exposure is self-selected**, and a plain regression of sales on exposure is biased **upward**: it credits the ad for enthusiasm that was already there. We want the *true* causal effect of an extra exposure so we can set budgets and frequency caps that don’t overspend.

1.0.1 The problem in one word: endogeneity

When the treatment (ad exposure) is **correlated with the hidden drivers of the outcome** (a customer’s unobserved intent), we say exposure is **endogenous**, and no amount of controlling for what we *measured* fixes it — the confounder (intent) is unmeasured. Notebook 05’s adjustment tools are powerless here.

1.0.2 The fix: an instrument

An **instrumental variable (IV)** is a third variable Z that nudges the treatment *without* affecting the outcome any other way. Here Z is a randomized **encouragement** — e.g. a serving-priority lottery that makes some customers *more likely* to be shown the ad. Because Z is randomized, the slice of exposure it drives is as-good-as-random, uncontaminated by intent. IV isolates that clean slice and reads the effect off it. A valid instrument needs three things:

- **Relevance** — Z actually moves exposure (this one we *can* test: the **first-stage F-statistic**; $F < 10$ is a “weak instrument” danger sign).
- **Exclusion** — Z affects sales *only through* exposure (the lottery itself sells nothing). Untestable from data — defended on design, and stress-tested in Step 6.
- **Exogeneity** — Z is independent of the hidden confounder (guaranteed here by randomization).

1.0.3 What IV recovers: a LATE

IV does **not** give the average effect over everyone. It gives the **LATE** (*Local Average Treatment Effect*) — the effect *for the “compliers,”* the customers whose exposure the encouragement actually

changed. Always-exposed loyalists and never-exposed sceptics are silent in an IV estimate; report the scope accordingly.

On real data. IV needs a genuine instrument, which is the hard part. Good marketing instruments: randomized **encouragement** / **intent-to-treat** designs (ship the nudge to a random half), ad-auction or delivery randomness, or a **cost shifter** for the price-elasticity problem in notebook 03. If you can randomize the *encouragement* (not the exposure itself), you have a clean instrument.

7-step contract.

1.1 2 · Simulate a ground truth

The **true** exposure->sales effect is **€15**. An unobserved **engagement** drives both exposure and sales (the confounding), so naive OLS overstates it. A randomized **encouragement** (the instrument) nudges exposure without touching sales directly.

TRUE exposure->sales effect €15 · first-stage: exposure 56%->77% (shift +21%),
F = 156

	encouragement	ad_exposure	sales
0	1.0	0.0	63.748992
1	1.0	1.0	64.636639
2	1.0	1.0	46.507627
3	0.0	1.0	75.599770
4	0.0	1.0	70.577754

1.2 3 · Identify — a valid instrument, and what IV recovers

Exposure is **endogenous** ($\text{Cov}(\text{exposure}, \varepsilon) \neq 0$), so OLS is biased. A valid instrument Z (encouragement) needs three things:

- **Relevance** — Z moves exposure. *Testable* (the first-stage F above; $F < 10$ = weak).
- **Exclusion** — Z affects sales *only through* exposure. *Untestable* — defended on design (a serving lottery sells nothing by itself). Stress-tested in step 6.
- **Exogeneity** — Z independent of the confounder. Holds by randomization.

What IV identifies — LATE (Imbens–Angrist): under monotonicity (no defiers), IV recovers the effect for **compliers** — customers the instrument moves — not everyone. CausalPy fits a Bayesian IV: a joint model of (exposure, sales) with correlated errors, *estimating* the endogeneity ρ rather than assuming it away.

1.3 4 · Estimate — OLS, reduced form, and Bayesian IV

Three regressions tell the whole story, and seeing them side by side demystifies what IV actually does:

- **OLS** — regress sales on exposure directly. This is the **biased** number, inflated by self-selection.
- **Reduced form** — regress sales on the *instrument* (encouragement -> sales). This captures only the effect that flows through the clean, randomized channel.

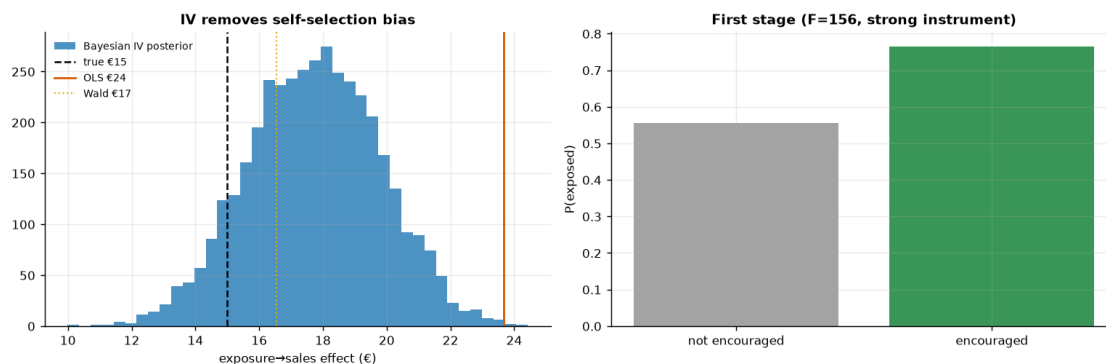
- **First stage** — regress exposure on the instrument (encouragement → exposure): how much the nudge moves exposure.
- **Wald / IV ratio** — reduced form ÷ first stage. The intuition: the instrument raised sales by *this* much and raised exposure by *that* much, so each unit of (instrument-driven) exposure is worth their ratio. That ratio *is* the causal effect. The **Bayesian IV** below is the same idea fit jointly with uncertainty (and it estimates the endogeneity ρ rather than assuming it away).

OLS (biased) €23.7
 Reduced form (Z→Y) €3.5 ÷ first stage (Z→D) 0.21 = Wald/IV ratio €16.5
 Bayesian IV €17.7 (true €15) 90% CI [€14.3, €21.1]
 IV convergence: max r-hat 1.010 - min ESS 903 - divergences 0

1.4 5 · Validate — IV removes the self-selection bias

Naive OLS should sit well *above* the truth; the Wald ratio and the Bayesian IV should both **remove that bias and land close to €15**, with the IV posterior’s 90% interval **covering the truth**. The gap between OLS and IV **is** the self-selection bias the instrument removes. (We give the Bayesian IV weakly-informative priors so its interval reflects genuine sampling uncertainty — roughly the frequentist Wald width — rather than the artificially tight band CausalPy’s default 2SLS-centred priors would produce.)

self-selection bias removed: OLS €23.7 → IV €17.7 (true €15)



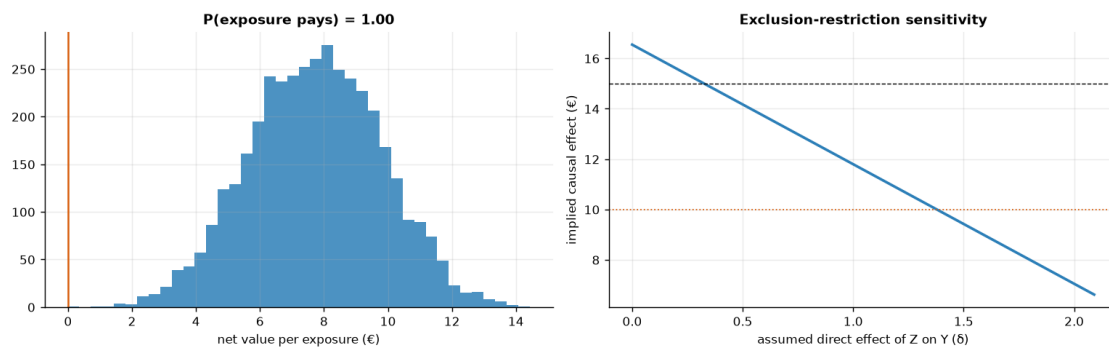
How to read this. *Left* — three vertical lines tell the whole story: **OLS (orange) sits well above the truth** (it credits the ad for pre-existing intent), while the **Bayesian IV posterior (blue) and the Wald ratio (gold) both drop back down toward the true €15**. The IV point carries a little finite-sample noise (a euro or two above 15), but its posterior — now honestly wide, thanks to weakly-informative priors — comfortably **covers €15**; the artificially tight default-prior band would have excluded it. The horizontal distance from the orange line to the blue posterior *is* the self-selection bias — the euros of “effect” that were never causal. If you budgeted on OLS you’d over-spend by exactly that gap and set frequency caps too high. *Right* — the first stage: the encouragement clearly moves exposure (a big jump in $P(\text{exposed})$), which is why the instrument is **strong** ($F \gg 10$). A weak first stage here would make the whole IV estimate unstable, so this panel is a precondition, not a footnote.

1.5 6 · Decide, in euros — and the exclusion-restriction stress test

Budget on the *causal* per-exposure value (IV), not the inflated OLS. But the exclusion restriction is untestable, so we ask: **what if the encouragement had a small direct effect on sales** (say the lottery email itself advertises)? We subtract a hypothetical direct effect δ from the reduced form and watch the implied causal estimate move — the honest bound on how much a leak would distort the number.

Causal net €7.7/exposure · P(pays) 1.00 → BUY exposure (to the complier margin)

Exclusion stress: the call survives unless the instrument's direct effect exceeds $\delta \sim 1.4$ (out of a €3.5 reduced form). If you trusted OLS (€24) you'd over-spend and set frequency caps too high.



1.6 7 · Caveats

- **LATE, not ATE.** IV speaks only for **compliers** — customers the encouragement moves. Always-exposed loyalists and never-exposed skeptics may respond differently; report the scope.
- **Weak instruments are dangerous.** A small first-stage (low F) inflates variance and bias. We checked F above; a weak instrument is worse than none.
- **Exclusion is untestable.** We stress-tested it in step 6; defend it on design, not data.
- **Monotonicity (no defiers).** IV's LATE interpretation assumes no one is pushed *away* from exposure by encouragement — usually plausible, worth stating.